5.1 Eigenvectors and Eigenvalues

The basic concepts presented here-eigenvectors and eigenvalues – are useful throughout pure and applied mathematics. Eigenvalues are also used to study difference equations and continuous dynamical systems. They provide critical information in engineering design, and they arise naturally in such fields as physics and chemistry.
http://www.youtube.com/watch?v=3mclp9QmCGs (Tacoma Bridge Video)

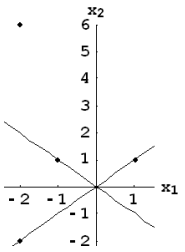
Definition: An eigenvector of an n x n matrix A is a nonzero vector $\mathbf{x}$ such that $A\mathbf{x} = \lambda\mathbf{x}$ for some scalar λ. A scalar λ is called an eigenvalue of A if there is a nontrivial solution $\mathbf{x}$ of $A\mathbf{x} = \lambda\mathbf{x}$; such an $\mathbf{x}$ is called an eigenvector corresponding to λ.

In other words, a nonzero vector $\mathbf{v}$ is an eigenvector of a matrix A iff the transformed vector $A\mathbf{x}$ points in the same or opposite direction of $\mathbf{v}$.

Ex 1: Let $A = \begin{bmatrix} 0 & -2 \\ -4 & 2 \end{bmatrix}$, $\mathbf{u} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, and $\mathbf{v} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$. Are $\mathbf{u}$ and $\mathbf{v}$ eigenvectors of A? Note: $A\mathbf{u} = -2\mathbf{u}$ but $A\mathbf{v} \neq \lambda\mathbf{u}$

If λ is an eigenvalue for a matrix A then the equation $(A - \lambda I)\mathbf{x} = \mathbf{0}$ has a nontrivial solution. The set of all solutions of this equation is the null space of the matrix $A - \lambda I$. So, this is a subspace of R^n and is called the eigenspace of A corresponding to λ.

Ex 2: Show that 4 is an eigenvalue of $A = \begin{bmatrix} 0 & -2 \\ -4 & 2 \end{bmatrix}$ and find the corresponding eigenvectors. (Hint: Scalar 4 is an eigenvalue of A if and only if $A\mathbf{x}=4\mathbf{x}$ has a nontrivial solution.)



Ex 3: Let $S = \begin{bmatrix} 2 & 0 & 0 \\ -1 & 3 & 1 \\ -1 & 1 & 3 \end{bmatrix}$. An eigenvalue of S is $\lambda = 2$. Find a basis for the corresponding eigenspace.

Ex 4: Let $A = \begin{bmatrix} 3 & 0 & -5 \\ \frac{1}{5} & -1 & 0 \\ 1 & 1 & -2 \end{bmatrix}$. The eigenvalues of A are $\lambda = 0, \pm\sqrt{2}$. Find a basis for the corresponding eigenspaces. (Hold for warm up for day #2)

Notice that while an eigenvalue λ might be zero, an eigenvector is never zero.

Ex 5: Suppose λ is an eigenvalue of A. Determine an eigenvalue of A^2 and A^3. In general, what is an eigenvalue of A^n?

Theorem 1 The eigenvalues of a triangular matrix are the entries on its main diagonal. (See proof on page 269).

Theorem 2: If $\mathbf{v}_1, \ldots \mathbf{v}_r$ are eigenvectors that correspond to distinct eigenvalues $\lambda_1, \ldots, \lambda_r$ of an n x n matrix A, then $\{ \mathbf{v}_1, \ldots \mathbf{v}_r \}$ is a linearly independent set. (See the proof on page 270).

5.2 The Characteristic Equation

How do we find the eigenvalues? We go back to the equation $A\mathbf{x} = \lambda\mathbf{x}$ where λ is a scalar.

Rearranging this equation, we see that this is a homogeneous system $A\mathbf{x} - \lambda\mathbf{x} = \mathbf{0}$

Factoring, we get $\mathbf{x}(A - \lambda) = \mathbf{0}$

If we insert I, we are subtracting two matrices. $\mathbf{x}(A - \lambda I) = \mathbf{0}$

What we want to do with this equation is to find the λ values which give a nontrivial solution.

These values of λ are called the characteristic values or eigenvalues.

If λ is an eigenvalue of A then the nontrivial solutions are the eigenvectors.

Since $A\mathbf{x} = 0$ has a nontrivial solution only if $\det A = 0$, we know that $\mathbf{x}(A - \lambda I) = 0$ has a nontrivial solution iff $\det(A - \lambda I) = 0$.

The equation $\det(A - \lambda I) = 0$ is called the characteristic equation.

characteristic polynomial: $\det(A - \lambda I)$

Ex 1: Given the following matrix, find

i. the characteristic equation
ii. the eigenvalues
iii. the eigenspaces corresponding to each eigenvalue

$$\begin{bmatrix} 0 & 1 \\ -6 & 5 \end{bmatrix}$$

For a 3 x 3 matrix or larger, recall that a determinant can be computed by cofactor expansion.

Ex 2: Find the eigenvalues and eigenvectors of $A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & -5 & 0 \\ 1 & 8 & 1 \end{bmatrix}$

Theorem – The Invertible Matrix Theorem (continued)

Let A be an n x n matrix. Then A is invertible if and only if:

s. The number 0 is not an eigenvalue of A.

t. $\det A \neq 0$

Ex3: Find the eigenvalues and eigenvectors of $A = \begin{bmatrix} 3 & 2 & 3 \\ 0 & 6 & 10 \\ 0 & 0 & 2 \end{bmatrix}$.

The (algebraic) multiplicity of an eigenvalue is its multiplicity as a root of the characteristic equation.

Ex 4: Find the characteristic polynomial of matrix A and then find all the eigenvalues and the algebraic multiplicity of each eigenvalue. Finally, find the eigenvectors for each eigenvalue $A = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 5 & 3 & 0 & 0 \\ 9 & 1 & 3 & 0 \\ 1 & 2 & 5 & -1 \end{bmatrix}$

Similarity. For n x n matrices A and B, we say that A is similar to B if there is an invertible matrix P such that

$P^{-1}AP = B$ or equivalently, $A = PBP^{-1}$.

Similarity is not the same as row equivalence.

Theorem 4: If n x n matrices A and B are similar, then they have the same characteristic polynomial and hence the same eigenvalues (with the same multiplicities).

Application to Markov Chains

Ex: Consider the migration matrix $M = \begin{bmatrix} .95 & .90 \\ .05 & .10 \end{bmatrix}$ and define $x_{k+1} = Mx_k$. It can be shown that $x_0, x_1, x_2, \ldots$ converges to a steady state vector $x = \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}$. Why?

The answer lies in examining the corresponding eigenvectors. First we find the eigenvalues:

$$\det(M - \lambda I) = \det\left(\begin{bmatrix} .95-\lambda & .90 \\ .05 & .10-\lambda \end{bmatrix}\right) = \lambda^2 - 1.05\lambda + 0.05$$

Solve $\lambda^2 - 1.05\lambda + 0.05 = 0$ By factoring $\lambda = 0.05,\ \lambda = 1$

It can be shown that the eigenspace corresponding to $\lambda = 1$ is span $\{\mathbf{v}_1\}$ where $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and the eigenspace corresponding to $\lambda = 0.05$ is span $\{\mathbf{v}_2\}$ where $\mathbf{v}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$. Note that $M\mathbf{v}_1 = \mathbf{v}_1$, and so $\begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}$ is our steady state vector.

Then for a given vector x_0,

$$x_0 = c_1 v_1 + c_2 v_2$$
$$x_1 = Mx_0 = M\left(c_1 v_1 + c_2 v_2\right)$$
$$= c_1 Mv_1 + c_2 Mv_2 = c_1 v_1 + c_2 (0.05) v_2$$
$$x_2 = Mx_1 = M\left(c_1 v_1 + c_2 (0.05) v_2\right)$$
$$= c_1 Mv_1 + c_2 (0.05) Mv_2 = c_1 v_1 + c_2 (0.05)^2 v_2$$

And in general, $x_k = c_1 v_1 + c_2 (0.05)^k v_2$

Thus, $\lim_{k\to\infty} x_k = \lim_{k\to\infty}\left(c_1 v_1 + c_2 (0.05)^k v_2\right) = c_1 v_1$ And this is the steady state when $c_1 = \frac{1}{2}$

5.3 Diagonalization

We've learned that the eigenvalues of a diagonal matrix are the entries on the main diagonal. Therefore, it is often useful to diagonalize a matrix.

It is very easy to find powers of diagonal matrices as illustrated in the following example.

The matrix D is a diagonal matrix (i.e. entries off the main diagonal are all zeros).

D^k is trivial to compute as the following example illustrates.

Ex 1: Let $D = \begin{bmatrix} 5 & 0 \\ 0 & 4 \end{bmatrix}$. Compute D^2 and D^3. In general, what is D^k, where k is a positive integer?

Suppose we wish to find the power of a non-diagonal matrix. The most efficient way of doing this is to diagonalize the given matrix.

In order to do this, we must look at similar matrices.

We learned in the last section that a matrix B is said to be similar to a matrix A if there exists an invertible matrix P such that $P^{-1}AP = B$ or equivalently, $A = PBP^{-1}$. In this case, we want our matrix B to be the diagonal matrix. So, the goal here is to develop a useful factorization $A = PDP^{-1}$, when A is an n x n matrix.

Procedure for Diagonalizing a Matrix

1. Find the characteristic equation for the matrix A and its eigenvalues, $\lambda_1, \lambda_2, \ldots, \lambda_n$
2. Find the corresponding eigenvectors of A, $\mathbf{p}_1, \mathbf{p}_2, \ldots, \mathbf{p}_n$
3. Form the matrix P having $\mathbf{p}_1, \mathbf{p}_2, \ldots, \mathbf{p}_n$ as its column
4. Form the matrix D with $\lambda_1, \lambda_2, \ldots, \lambda_n$ on the diagonal corresponding to the same order as $\mathbf{p}_1, \mathbf{p}_2, \ldots, \mathbf{p}_n$

5. Check that $A = PDP^{-1}$

Ex 2: Given $A = \begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix}$, find matrices P and D such that $A = PDP^{-1}$.

Ex 3: Given $A = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 2 & 1 \\ -1 & 0 & 1 \end{bmatrix}$, find matrices P and D such that $A = PDP^{-1}$.

Ex 4: Let $A = \begin{bmatrix} 6 & -1 \\ 2 & 3 \end{bmatrix}$. Find a formula for A^k given that $A = PDP^{-1}$ where $P = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$, $D = \begin{bmatrix} 5 & 0 \\ 0 & 4 \end{bmatrix}$ and $P^{-1} = \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}$.

Note: $A^k = PD^kP^{-1}$

Ex 5: Compute A^{10} for the matrix $A = \begin{bmatrix} 1 & -2 \\ 3 & -4 \end{bmatrix}$

A square matrix A is said to be diagonalizable if A is similar to a diagonal matrix, i.e., if $A = PDP^{-1}$ where P is invertible and D is a diagonal matrix.

When is A diagonalizable? (The answer lies in examining the eigenvalues and eigenvectors of A.)

Theorem 5: The Diagonalization Theorem: An n x n matrix A is diagonalizable if and only if A has n linearly independent eigenvectors.

If fact, $A = PDP^{-1}$, with D a diagonal matrix, if and only if the columns of P are n linearly independent eigenvectors of A. In this case, the diagonal entries of D are eigenvalues of A that correspond, respectively, to the eigenvectors in P.

Ex 6: Why is $A = \begin{bmatrix} 2 & 0 & 0 \\ 2 & 6 & 0 \\ 3 & 2 & 1 \end{bmatrix}$ diagonalizable?

Theorem 6: An n x n matrix with n distinct eigenvalues is diagonalizable.

Ex 7: Show that $A = \begin{bmatrix} 2 & 0 \\ 1 & 2 \end{bmatrix}$ is not diagonalizable.

Ex 8: Diagonalize the following matrix, if possible. $A = \begin{bmatrix} -2 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 24 & -12 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}$

Theorem 7: Let A be an n x n matrix whose distinct eigenvalues are $\lambda_1, \lambda_2, \ldots, \lambda_p$.

a. For $1 \le k \le p$, the dimension of the eigenspace for λ_k is less than or equal to the multiplicity of the eigenvalue λ_k.

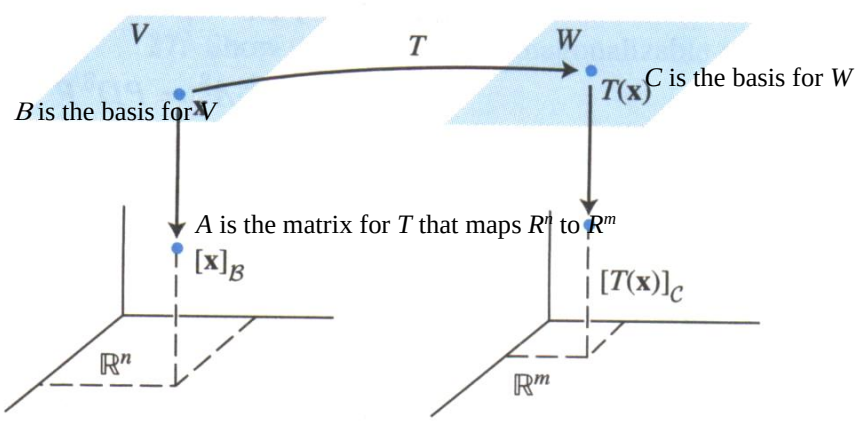
b. The matrix *A* is diagonalizable if and only if the sum of the dimensions of the distinct eigenspaces equals *n*, and this happens if and only if the dimension of the eigenspace for each λ_k equals the multiplicity of λ_k.

c. If *A* is diagonalizable and β_k is a basis for the eigenspace corresponding to λ_k for each *k*, then the total collection of vectors in the sets $\beta_1, \ldots, \beta_p$ forms an eigenvector basis for R^n.

5.4 Eigenvectors and Linear Transformations

Our goal is to understand the matrix factorization
$A = PDP^{-1}$ as a statement about linear transformations.

Recall from 1.9 that any linear transformation *T* from R^n to R^m can be implemented by left-multiplication by a matrix *A*.
There is a diagram on p 328 illustrating
the matrix of a linear transformation



The Matrix of a Linear Transformation

Let *V* be an *n*-dimensional vector space, *W* an m-dimensional vector space, and *T* any linear transformation from *V* to *W*. To associate a matrix with *T*, choose (ordered) bases *B* and *C* for *V* and *W*, respectively.

Let $B = \{\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_n\}$.
We want $A[\mathbf{b}_1]_B = [T(\mathbf{b}_1)]_C$,
$A[\mathbf{b}_2]_B = [T(\mathbf{b}_2)]_C$, …etc.

$[T(\mathbf{x})]_C = M[\mathbf{x}]_B$ where

$M = [\,[T(\mathbf{b}_1)]_C\,[T(\mathbf{b}_2)]_C \ldots [T(\mathbf{b}_n)]_C\,]$

Where *M* is the matrix for *T* relative to *B* and *C*

M is a matrix representation of *T*, called the matrix for *T* relative to the bases *B* and *C*.

The action of *T* on **x** may be viewed as left multiplication by *M*.

Algorithm for finding *B* matrix of *T: V→W*

1. Compute the images of the basis vectors $T(\mathbf{b}_1)$, $T(\mathbf{b}_2)$, …, $T(\mathbf{b}_n)$
2. Convert these images into *B* coordinate vectors $[T(\mathbf{b}_1)]_B, [T(\mathbf{b}_2)]_B, \ldots, [T(\mathbf{b}_n)]_B$
3. Place *B* vectors into columns of $[T]_B$

Ex 1: Let $D = \{\mathbf{d}_1, \mathbf{d}_2\}$ and $B = \{\mathbf{b}_1, \mathbf{b}_2\}$ be bases for vector spaces V and W, respectively. Let $T: V \to W$ be a linear transformation with the property that

$$T(\mathbf{d}_1) = 2\mathbf{b}_1 - 3\mathbf{b}_2 ,$$
$$T(\mathbf{d}_2) = -4\mathbf{b}_1 + 5\mathbf{b}_2$$

Find the matrix for T relative to D and B.

Ex 2: Let $B = \{\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\}$ and $D = \{\mathbf{d}_1, \mathbf{d}_2\}$ and be bases for vector spaces V and W, respectively. Let $T: V \to W$ be a linear transformation with the property that

$$T(\mathbf{b}_1) = 3\mathbf{d}_1 - 5\mathbf{d}_2$$
$$T(\mathbf{b}_2) = -\mathbf{d}_1 + 6\mathbf{d}_2$$
$$T(\mathbf{b}_3) = 4\mathbf{d}_2$$

Find the matrix for T relative to B and D.

If B and C are bases for the same space V and if T is the identity transformation $T(\mathbf{x}) = \mathbf{x}$ for $\mathbf{x}$ in V, then the matrix M is just a change-of-coordinates matrix (see section 4.7).

Linear Transformations from V into V

In the common case when W is the same as V and the basis C is the same as B, the matrix M (in the earlier equation) is called the matrix for T relative to B, or simply the B-matrix for T, and is denoted by $[T]_B$

The B-matrix for $T: V \to V$ satisfies $[T(\mathbf{x})]_B = [T]_B[\mathbf{x}]_B$, for all $\mathbf{x}$ in V.

Ex 3: Let $T : P_2 \to P_4$ be the transformation that maps a polynomial $\mathbf{p}(t)$ into the polynomial $\mathbf{p}(t) + t^2\,\mathbf{p}(t)$

a. Find the image of $\mathbf{p}(t) = 2 - t + t^2$

b. Show that T is a linear transformation.

c. Find the matrix for T relative to the bases $\{1, t, t^2\}$ and $\{1, t, t^2, t^3, t^4\}$.

Linear Transformations on R^n

In an applied problem involving R^n, T usually appears first as a matrix transformation, $\mathbf{x} \to A\mathbf{x}$.

If A is diagonalizable, then there is a basis B for R^n consisting of eigenvectors of A.

Diagonalizing a matrix A amounts to finding a diagonal matrix representation of , $\mathbf{x} \to A\mathbf{x}$

Theorem 8 – Diagonal Matrix Representation

Suppose $A = PDP^{-1}$, where D is a diagonal n x n matrix. If B is the basis for R^n formed from the columns of P, then D is the B-matrix for the transformation $\mathbf{x} \to A\mathbf{x}$.

Similarity of Matrix Representations

If A is similar to a matrix C, with $A = PDP^{-1}$, then C is the B-matrix for the transformation $\mathbf{x} \to A\mathbf{x}$ when the basis V is formed from the columns of P.

Conversely, if $T: R^n \to R^n$ is defined by $T(\mathbf{x}) = A\mathbf{x}$, and if V is any basis for R^n, then the B-matrix for T is similar to A.

If P is the matrix whose columns come from the vectors in B, then $[T]_B = P^{-1}AP$.

The set of all matrices similar to a matrix A coincides with the set of all matrix representations of the transformation $\mathbf{x} \to A\mathbf{x}$.

Ex 4: Find the B-matrix for the transformation $\mathbf{x} \to A\mathbf{x}$, when $B = \{\mathbf{b}_1, \mathbf{b}_2\}$ $A = \begin{bmatrix} -1 & 4 \\ -2 & 3 \end{bmatrix}$, $\mathbf{b}_1 = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$, $\mathbf{b}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$,

Ex 5: Define $T: R^2 \to R^2$ by $T(\mathbf{x}) = A\mathbf{x}$. Find a basis B for R^2 with the property that $[T]_B$ is diagonal. $A = \begin{bmatrix} 5 & -3 \\ -7 & 1 \end{bmatrix}$

Ex 6: Find the B-matrix for the transformation $\mathbf{x} \to A\mathbf{x}$ when $B = \{\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\}$

$$A = \begin{bmatrix} -14 & 4 & -14 \\ -33 & 9 & -31 \\ 11 & -4 & 11 \end{bmatrix}, b_1 = \begin{bmatrix} -1 \\ -2 \\ 1 \end{bmatrix}, b_2 = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}, b_3 = \begin{bmatrix} -1 \\ -2 \\ 0 \end{bmatrix}$$